

FROM TRADING IDEA TO MT5 EA WITH AI

A Practical Step-by-Step Guide to Turning a Trading Idea Into a Working MT5 Expert Advisor

How to use ChatGPT and Claude to transform a simple trading idea into an MT5 Expert Advisor — without writing the code yourself.

PREPARED FOR LEARNERS

Worked example: [Previous-Day High/Low Breakout EA \(MQL5\)](#)

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1 INTRODUCTION

You do not need to be an experienced programmer to begin turning your trading ideas into automated trading systems. Modern AI tools can help you move from an idea written in plain English to a working Expert Advisor (EA) for MetaTrader 5.

KEY PRINCIPLE: AI does not replace the thinking required to design a trading strategy. You still need to explain, precisely, what you want the robot to do.

The process taught in this guide is therefore not "ask AI to make me a trading robot." Instead, we follow a structured process:

Trading Idea !' Specification !' Review !' Coding Prompt !' MQL5 Code !' MetaEditor !' Compilation !' Backtest !' Verification

2 THE AI TRADING ROBOT WORKFLOW

- ChatGPT — the strategy partner: developing the idea, asking clarifying questions, turning idea into technical specification, creating coding prompt for Claude.
- Claude — the coding partner: turning specification into MQL5 code, producing the complete EA, fixing compilation errors, producing fresh complete code versions.
- MetaEditor — the build environment: where we create the EA file, paste MQL5 code, compile it, and identify compiler errors.
- MT5 Strategy Tester — the proving ground: run EA against historical tick data, verify trade logic, and examine drawdown/execution.

3 PHASE 1 — STRATEGY DESIGN: WORKED EXAMPLE

Our worked example is the Previous-Day High/Low Breakout EA:

1. At the beginning of each trading day, identify yesterday's High and Low from completed D1 bar.
2. Buy when price genuinely crosses above yesterday's High.
3. Sell when price genuinely crosses below yesterday's Low.
4. Use fixed-pip Stop Loss and Take Profit.
5. Allow configurable number of trades per direction per day (e.g., max 3 Buys, max 3 Sells).
6. Allow Buy and Sell trades independently of one another.
7. Filter trades by maximum allowable spread.
8. Enter immediately upon crossing the level (no candle-close delay).
9. Reset levels, baseline, and counters at broker 00:00.

4 & 5 PHASES 2 & 3 — CODING & METAEDITOR COMPILATION

1. Request complete code from Claude using structured prompt.
2. In MetaEditor: Tools !' MetaQuotes Language Editor !' File !' New !' Expert Advisor (template).
3. Name file PrevDayBreakoutEA.mq5, delete template code, paste Claude's code, press Compile (F7).
4. Goal: "0 errors, 0 warnings" in MetaEditor Errors tab.

6 & 7 PHASES 4 & 5 — DEBUGGING & STRATEGY TESTER

If errors occur, never guess. Copy the full compiler error message, return to Claude, and prompt:

"The EA does not compile in MetaEditor. Here are the compilation errors: [paste errors]. Please fix all errors and provide the complete fresh MQL5 code."

Once compiled with 0 errors, open MT5 Strategy Tester (Ctrl+R / View !' Strategy Tester):

- Select PrevDayBreakoutEA.ex5
- Select Symbol (e.g. EURUSD, GBPUSD)
- Timeframe: H1 (or your choice)
- Enable Visual Mode to inspect trades step by step on chart.

Final Verification Checklist

- ☐ Is the trading idea clearly defined in plain English?
- ☐ Did ChatGPT turn the idea into a detailed specification with SL/TP/limits?
- ☐ Did you personally review, modify, or approve the specification?
- ☐ Did Claude generate the complete, compilable MQL5 source code?
- ☐ Does MetaEditor compile with 0 errors?
- ☐ Does the EA open trades only upon genuine breakout crossing?
- ☐ Are Stop Loss and Take Profit placed at exact pip distances?
- ☐ Does the daily trade counter reset properly on a new trading day?

Appendix A — Complete MQL5 Source Code: PrevDayBreakoutEA.mq5

This is the complete, working MQL5 Expert Advisor implementing the Previous-Day High/Low Breakout strategy. Ready to compile directly in MetaEditor:

```
//+-----+
//|                                     PrevDayBreakoutEA.mq5 |
//|               Previous Day High/Low Breakout Expert Advisor |
//|
//| Strategy:
//| - At start of each broker day, read completed previous D1 candle
//|   and store High/Low as fixed breakout levels for current day.
//| - Open a BUY the instant price crosses above Previous Day High.
//| - Open a SELL the instant price crosses below Previous Day Low.
//| - Independent configurable max trades per direction per day.
//| - Fixed pip SL/TP, fixed lot size, max spread filter.
//+-----+

#property copyright "Generated EA"
#property version "1.00"
#property strict

#include <Trade\Trade.mqh>

//=====
// INPUTS
//=====
input double LotSize           = 0.10;      // Fixed lot size
input int    MaxBuyTradesPerDay = 3;        // Max BUY trades per day
input int    MaxSellTradesPerDay = 3;       // Max SELL trades per day
input double StopLossPips      = 20;       // Stop Loss in pips
input double TakeProfitPips    = 40;       // Take Profit in pips
input double MaxSpreadPips     = 3.0;      // Max allowed spread in pips
input long   MagicNumber       = 20260917; // Unique magic number

//=====
// GLOBALS
//=====
CTrade   g_trade;
double   g_pipSize = 0.0;
double   g_point   = 0.0;
int       g_digits  = 0;
datetime g_currentDayStart = 0;
double   g_prevDayHigh = 0.0;
double   g_prevDayLow  = 0.0;
int       g_buyCountToday = 0;
int       g_sellCountToday = 0;
double   g_prevBid = 0.0;
double   g_prevAsk = 0.0;
bool      g_priceBaselineSet = false;
string    g_objPrefix = "";
string    g_statusText = "Initializing";

double CalcPipSize()
{
    int digits = (int)SymbolInfoInteger(_Symbol, SYMBOL_DIGITS);
    double point = SymbolInfoDouble(_Symbol, SYMBOL_POINT);
    if(digits == 3 || digits == 5) return point * 10.0;
    return point;
}

datetime DayStart(datetime t)
{
    MqlDateTime mt;
    TimeToStruct(t, mt);
    mt.hour = 0; mt.min = 0; mt.sec = 0;
    return StructToTime(mt);
}

double ValidateLot(double lots)
{
    double minLot = SymbolInfoDouble(_Symbol, SYMBOL_VOLUME_MIN);
    double maxLot = SymbolInfoDouble(_Symbol, SYMBOL_VOLUME_MAX);
    double lotStep = SymbolInfoDouble(_Symbol, SYMBOL_VOLUME_STEP);
    if(lotStep <= 0.0) lotStep = 0.01;
    double steps = MathRound(lots / lotStep);
    double normalized = steps * lotStep;
    if(normalized < minLot) normalized = minLot;
    if(normalized > maxLot) normalized = maxLot;
    return NormalizeDouble(normalized, 2);
}
```

```

double GetSpreadPips()
{
    double bid = SymbolInfoDouble(_Symbol, SYMBOL_BID);
    double ask = SymbolInfoDouble(_Symbol, SYMBOL_ASK);
    if(g_pipSize <= 0.0) return 0.0;
    return (ask - bid) / g_pipSize;
}

void RecalculateDailyCounts()
{
    g_buyCountToday = 0;
    g_sellCountToday = 0;
    datetime fromTime = g_currentDayStart;
    datetime toTime = TimeTradeServer() + 60;
    if(!HistorySelect(fromTime, toTime)) return;
    int total = HistoryDealsTotal();
    for(int i = 0; i < total; i++)
    {
        ulong dealTicket = HistoryDealGetTicket(i);
        if(dealTicket == 0) continue;
        if(HistoryDealGetInteger(dealTicket, DEAL_MAGIC) != MagicNumber) continue;
        if(HistoryDealGetString(dealTicket, DEAL_SYMBOL) != _Symbol) continue;
        if(HistoryDealGetInteger(dealTicket, DEAL_ENTRY) != DEAL_ENTRY_IN) continue;
        if((datetime)HistoryDealGetInteger(dealTicket, DEAL_TIME) < g_currentDayStart) continue;

        long dealType = HistoryDealGetInteger(dealTicket, DEAL_TYPE);
        if(dealType == DEAL_TYPE_BUY) g_buyCountToday++;
        else if(dealType == DEAL_TYPE_SELL) g_sellCountToday++;
    }
}

bool LoadPreviousDayLevels()
{
    double highs[], lows[];
    ArraySetAsSeries(highs, true);
    ArraySetAsSeries(lows, true);
    if(Bars(_Symbol, PERIOD_D1) < 2) return false;
    if(CopyHigh(_Symbol, PERIOD_D1, 1, 1, highs) <= 0) return false;
    if(CopyLow(_Symbol, PERIOD_D1, 1, 1, lows) <= 0) return false;
    g_prevDayHigh = highs[0];
    g_prevDayLow = lows[0];
    return true;
}

void CheckNewDay()
{
    datetime todayStart = DayStart(TimeTradeServer());
    if(todayStart == g_currentDayStart && g_currentDayStart != 0) return;
    g_currentDayStart = todayStart;
    if(!LoadPreviousDayLevels()) { g_statusText = "Waiting for D1 data"; return; }
    RecalculateDailyCounts();
    g_priceBaselineSet = false;
    g_statusText = "Running";
}

void OpenBuyTrade(double entryPrice)
{
    double lots = ValidateLot(LotSize);
    double sl = (StopLossPips > 0) ? NormalizeDouble(entryPrice - StopLossPips * g_pipSize, g_digits) : 0.0;
    double tp = (TakeProfitPips > 0) ? NormalizeDouble(entryPrice + TakeProfitPips * g_pipSize, g_digits) :
0.0;
    g_trade.SetExpertMagicNumber(MagicNumber);
    if(g_trade.Buy(lots, _Symbol, 0.0, sl, tp, "PrevDayBreakout-Buy"))
        g_buyCountToday++;
}

void OpenSellTrade(double entryPrice)
{
    double lots = ValidateLot(LotSize);
    double sl = (StopLossPips > 0) ? NormalizeDouble(entryPrice + StopLossPips * g_pipSize, g_digits) : 0.0;
    double tp = (TakeProfitPips > 0) ? NormalizeDouble(entryPrice - TakeProfitPips * g_pipSize, g_digits) :
0.0;
    g_trade.SetExpertMagicNumber(MagicNumber);
    if(g_trade.Sell(lots, _Symbol, 0.0, sl, tp, "PrevDayBreakout-Sell"))
        g_sellCountToday++;
}

void CheckBreakouts()
{
    if(g_prevDayHigh <= 0.0 || g_prevDayLow <= 0.0) return;
    double bid = SymbolInfoDouble(_Symbol, SYMBOL_BID);
    double ask = SymbolInfoDouble(_Symbol, SYMBOL_ASK);
    if(bid <= 0.0 || ask <= 0.0) return;
}

```

```

if(!g_priceBaselineSet)
{
    g_prevBid = bid;
    g_prevAsk = ask;
    g_priceBaselineSet = true;
    return;
}

bool spreadOk = (GetSpreadPips() <= MaxSpreadPips);
bool buyCross = (g_prevAsk <= g_prevDayHigh && ask > g_prevDayHigh);
bool sellCross = (g_prevBid >= g_prevDayLow && bid < g_prevDayLow);

if(buyCross && g_buyCountToday < MaxBuyTradesPerDay && spreadOk)
    OpenBuyTrade(ask);

if(sellCross && g_sellCountToday < MaxSellTradesPerDay && spreadOk)
    OpenSellTrade(bid);

g_prevBid = bid;
g_prevAsk = ask;
}

int OnInit()
{
    g_digits = (int)SymbolInfoInteger(_Symbol, SYMBOL_DIGITS);
    g_point = SymbolInfoDouble(_Symbol, SYMBOL_POINT);
    g_pipSize = CalcPipSize();
    g_objPrefix = "PDBO_" + IntegerToString(MagicNumber) + "_";
    g_trade.SetExpertMagicNumber(MagicNumber);
    g_trade.SetDeviationInPoints(10);
    g_trade.SetTypeFilling(ORDER_FILLING_FOK);

    if(LotSize <= 0.0) return(INIT_PARAMETERS_INCORRECT);
    g_currentDayStart = 0;
    g_priceBaselineSet = false;
    CheckNewDay();
    return(INIT_SUCCEEDED);
}

void OnDeinit(const int reason) {}

void OnTick()
{
    CheckNewDay();
    CheckBreakouts();
}

```


